HOW TO USE: QUICK START & WORKFLOW

Step-by-step guide to finding roots using bisection.py (v2.0)

1. Launching the Application

• Open your terminal and run the script with Python:

```
$ python bisection.py
```

2. The 4-Step Input Process

Step A: Enter Function f(x)

Type your mathematical expression in terms of 'x'.
Examples: 2x + sin(x), x^2 - 2, exp(x) - 3x, x^3 - 7x + 1
• Use '^' for powers. Implicit multiplication works naturally (e.g. 2x, 2sin(x)).

Step B: Enter Interval [a, b]

Enter two comma-separated numbers: a,b (e.g. -1,10 or 0,1).
• The app enforces Bolzano's Theorem: f(a) and f(b) MUST have opposite signs.
• If an endpoint is already an exact root, it finishes immediately!

Step C: Enter Relative Error Tolerance

Specify the stopping tolerance for Relative Approximate Error (ea <= tol).
• Examples: 0.000001 or 0.0001% (Press Enter to use default: 0.0001% = 1e-6).
• The method halts automatically once the relative shift falls below this threshold.

Step D: Optional Known True Root (Analysis Tool)

Enter known exact root if you want to benchmark accuracy (e.g. 0, sqrt(2)) [Enter = skip].
• If provided, the app computes both Absolute Exact Error (Et) and
  Relative Exact Error (et) in the final summary report!

3. Iteration Table & Interactive Session Loop

• Prints a clean 6-column iteration table (tracking bounds, midpoint, Ea, and ea).

• Final summary reports original function, range, tolerance, root, and error types.

• Automatic prompt asks whether to generate and display the 2-panel graphical plot!

```
Press Enter to solve another (or 'q' to quit) | Enter 'n' to skip plot generation
```

SYNTAX GUIDE & THE 4 NUMERICAL ERRORS

Expression reference and error classification

Mathematical Operators & Functions

Category	Syntax Example	Interpretation / Description
Powers	x^2 - 2, x^3 + 4	Caret '^' automatically converts to Python's '**' operator.
Implicit Multi.	2x, 2sin(x), 3(x+1)	Numbers/variables preceding functions/brackets auto-insert '*'.
Trigonometric	sin(x), cos(2x), tan(x)	Standard radian trigonometry functions.
Exponential	exp(x), exp(-x^2)	Natural exponential e^x.
Logarithmic	log(x), log10(x)	log(x) is natural ln(x). log10(x) is common base-10 log.
Constants	pi, e, tau	pi ~ 3.14159, e ~ 2.71828.
Scientific Not.	1e-5, 2.5e3	Preserved cleanly without unwanted multiplication asterisks.

Understanding the 4 Types of Errors

- **1. Relative Approx Error (ea) [STOPPING CRITERION]:** `|c_curr - c_prev| / |c_curr|`

  The sole convergence stopping criterion! Halts iterations when ea <= tolerance.
  Displayed in table column 'Rel Approx (ea)'.

- **2. Absolute Approx Error (Ea) [ANALYSIS TOOL]:** `|c_curr - c_prev|`

  Measures the absolute step shift between consecutive midpoints.
  Displayed in table column 'Approx Err (Ea)'.

- **3. Absolute Exact Error (Et) [ANALYSIS TOOL]:** `|x_true - c|`

  The true physical distance to the exact root. Reported in the final summary if true root is provided.

- **4. Relative Exact Error (et) [ANALYSIS TOOL]:** `|x_true - c| / |x_true|`

  The exact error normalized by the true root. (Undefined if true root is 0 due to division by zero).

Relative Error as the Standard Stopping Rule:

Relative approximate error ensures that stopping is scale-invariant (e.g. 0.0001% accuracy works equally well whether the root is 1.414 or 14,000). The other 3 errors give you full analytical insight!

GRAPHICAL PLOTS & SUMMARY CARD

2-panel visualization, banners, and results card

2-Panel Graphical Convergence Plot (PNG Export)

After each run, the application can generate a publication-quality 2-panel chart (bisection_plot.png):

- **Left Panel: Root Localization:**

  Plots the continuous function curve f(x), shades the initial bracket [a, b], marks the endpoints a and b, displays intermediate midpoints c_k, and highlights the root with a star.

- **Right Panel: Error Decay Curve:**

  Plots Relative Approximate Error (ea) and Absolute Approximate Error (Ea) vs. iteration on a logarithmic scale against the dotted red tolerance line, annotating the stopping step.

- **Interactive GUI & File Export:**

  Automatically saves high-resolution bisection_plot.png (200 DPI) and displays an interactive desktop window that can be zoomed, panned, or saved.

Convergence Summary with Problem Context

```
+=============================================================================================================+
|                                3. CONVERGENCE SUMMARY & ERROR ANALYSIS                                      |
+=============================================================================================================+
  Problem Specification:
    * Function f(x)       : x^3 - 7x + 1
    * Initial Range [a, b]: [0, 1]
    * Error Tolerance     : 0.001 (0.1%)
    * Stopping Criterion  : Relative Approximate Error (ea <= tolerance)

  Solution Results:
    * Approximate Root (c): 0.1431884766
    * Function Value f(c) : +6.164488e-04  (effectively zero)
    * Iterations Required : 13
    * Termination Reason  : Reached relative tolerance or exact root

  Error Analysis (4 Numerical Errors):
    1. Relative Approx Error (ea) [Stopping Metric]: 8.5251e-04 (8.5251e-02%)
    2. Absolute Approx Error (Ea)                  : 1.220703e-04
    3. Absolute Exact Error  (Et = |x_true - c|)   : 8.884344e-05
    4. Relative Exact Error  (et = |x_true - c|/|x_t|): 6.2008e-04 (6.2008e-02%)
+=============================================================================================================+
```

Visual Verification Made Simple:

The combination of standardized banners, adaptive terminal tables, and high-resolution 2-panel plots gives you total confidence in your numerical solution from both an algebraic and visual perspective!